NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

Department of Computer Science and Engineering

Title of the Project

A Blended Deep Learning Approach for Hidden Prediction of Smoking Status to Reduce the Risk of Cardiovascular Disease

PRESENTED BY

DUGGI SAMPATH	(22471A0518)
CHIRUMALA CHARAN TEJA	(22471A0513)
GARIKAPUDI PRUDHVIRAJ	(22471A0522)

Under the Guidance of,

Gaddam Saranya Ass.prof,

Designation,

Department of Computer Science and Engineering,

Narasaraopeta Engineering College (Autonomous),

Narasaraopet- 522 601.

ABSTRACT

- Abstract of the project :
- Focus: Detection of **hidden or undeclared smokers** to reduce **cardiovascular disease (CVD) risk**.
- Dataset: Large **biosignal dataset** with multiple physiological features.
- Preprocessing: **Data cleaning**, **scaling**, and **class imbalance handling** using **Borderline-SMOTE**.
- Feature Selection: **Random Forest** used to identify most influential features.
- Model: Proposed **blended deep learning model (PBM)** combining **ESN** and **GoogleNet**, with **AlexNet** as meta-learner.
- Evaluation: Performance measured using **10-fold cross-validation** and standard classification metrics.
- Key Result: Achieved **high prediction accuracy** with **explainability** provided through **SHAP** analysis.

INTRODUCTION

- Brief introduction to the project

Smoking is a major contributor to **cardiovascular diseases (CVD)**. Many individuals **hide or under-report** their smoking habits.

Physiological and clinical data can reveal **subtle indicators** of smoking status. The project uses **machine learning and deep learning** to detect hidden smokers.

- Motivation behind the project

Hidden smokers create **misleading medical records**, affecting treatment accuracy.

Increasing availability of **biosignal datasets** enables automated prediction.

- Importance and relevance

Supports **public health programs** aimed at smoking cessation and awareness. Enhances **clinical decision-making** with explainable model output (e.g., SHAP).

NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

LITERATURE SURVEY

No	Title	Author	Journal Name & Year	Methodology Adapted	Key Findings	Gaps
1	Font size = 14		URL of the paper			
2						
3						
4						
5						

LITERATURE SURVEY

- Previous research focused on detecting smoking behavior using **wearable signals** and **deep learning models**.
- Some studies used **LSTM / RNN** for gesture or biosignal-based smoking detection.
- Others applied **CNN-based models** for image or signal pattern recognition.
- However, most existing models used **only one type of feature representation** (either temporal or spatial).
- **Model interpretability** was limited in earlier works, reducing clinical acceptance.
- **Our proposed PBM model** addresses these gaps by **combining ESN + GoogleNet + AlexNet** with **SHAP explainability**.

RESEARCH GAPS

Aspect	Existing Methods	Proposed PBM Model
Data Reliability	Self-reported smoking data	Physiological & biosignal-based
Class Imbalance	Often ignored or basic SMOTE	Borderline-SMOTE applied
Feature Selection	Limited or none	Random Forest selected features
Model Type	Single neural model	Blended ESN + GoogleNet + AlexNet

- Smoking status is often hidden or misreported, making traditional assessments unreliable.
- Existing smoking detection models do not handle class imbalance effectively, leading to biased prediction.
- Many previous works use single-model architectures, limiting feature learning capability.
- Earlier research focuses on only one type of feature pattern (either temporal or spatial), reducing accuracy.
- Prior models lack interpretability, making it difficult for doctors to trust and use predictions.
- There is a need for a hybrid model that can combine multiple feature representations and provide explainable insights.

PROBLEM STATEMENT

- Many individuals **hide or under-report** their smoking habits in medical examinations.
- This leads to **incorrect cardiovascular risk assessment** and delayed clinical intervention.
- Traditional detection methods rely on **self-reported data**, which is often unreliable.
- There is a need for an **automated and accurate system** to identify **hidden smokers** using physiological data.
- The challenge is to **handle class imbalance**, extract meaningful features, and ensure **model interpretability** for clinical use.

OBJECTIVES

- To **preprocess the biosignal dataset** using Robust Scaling, duplicate removal, and **Borderline-SMOTE** to handle class imbalance.
- To **select the most relevant features** using the **Random Forest** feature importance method.
- To **develop a blended deep learning model (PBM)** that combines **ESN** and **GoogleNet** as base learners, with **AlexNet** as the meta-classifier.
- To **evaluate model performance** using 10-fold cross-validation and provide **explainable predictions** using **SHAP analysis**.

BLOCK DIAGRAM OR FLOW DIAGRAM

METHODOLOGY

- Data Preprocessing
  - Dataset contains **physiological and clinical biosignal features**
  - **Robust Scaling** applied to minimize effect of outliers
  - **Missing target entries removed** and duplicates cleaned
- Feature Selection
  - **Random Forest** used to **rank and select important features**
  - Reduces model complexity and enhances prediction accuracy
- Model Architecture (PBM)
  - **ESN** extracts **temporal patterns** from input biosignals
  - **GoogleNet** captures **spatial and multi-scale feature structure**

Model Explainability

SHAP analysis used to interpret feature contribution

Supports clinical trust and decision transparency

Evaluation

10-fold Cross-Validation performed to ensure robustness

Performance assessed using Accuracy, Precision, Recall, and F1-score

IMPLEMENTATION

Software & Tools Used

- Programming Language: **Python**
- Deep Learning Frameworks: **TensorFlow / PyTorch**
- Additional Libraries: **Scikit-learn**, **Imbalanced-Learn**, **SHAP**

Hardware Requirements

- System with **GPU support** recommended for faster model training
- Minimum: **8GB RAM**, **Intel i5 / Ryzen 5** or above

Implementation Steps

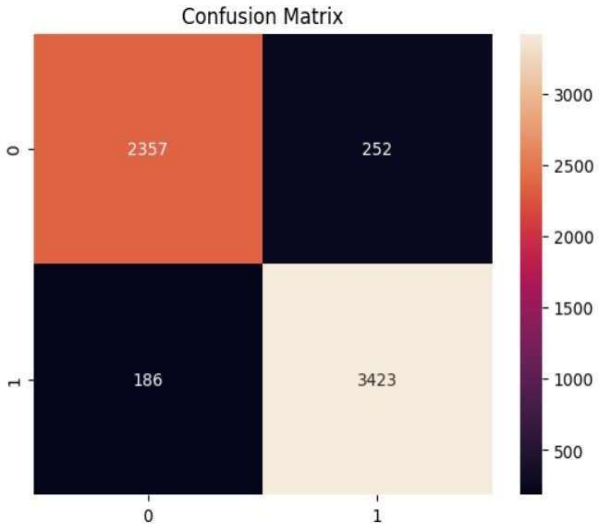
- Loaded and cleaned the **biosignal dataset**
- Performed **Robust Scaling** and **duplicate removal**
- Applied **Borderline-SMOTE** to balance class distribution.
- Extracted key features using **Random Forest**
- Trained **ESN** and **GoogleNet** as base learners
- Combined outputs and trained **AlexNet** as the meta-classifier
- Evaluated model using **10-fold cross-validation**

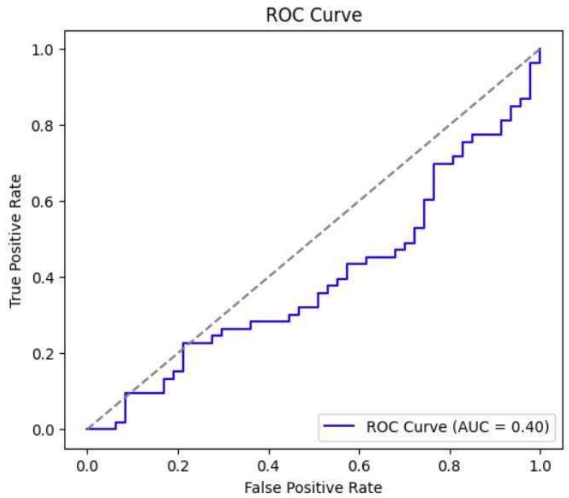
Challenges & Solutions

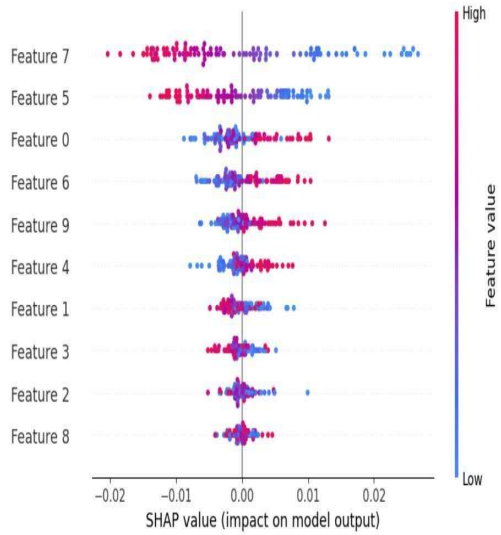
- **Class imbalance** → Solved using **Borderline-SMOTE**
- **High feature dimensionality** → Reduced using **Random Forest selection**
- **Need for interpretability** → Addressed using **SHAP explainability**

NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

RESULTS & ANALYSIS







- The Proposed Blended Model (PBM) was evaluated using **10-fold cross-validation.**
- Achieved an overall **accuracy of ~92%** in predicting smoking status.
- Both classes (**Smoker** and **Non-Smoker**) showed **high precision and recall**, confirming reliability.
- **Confusion matrix** indicates a **low number of misclassifications**.
- **SHAP explainability** highlights key contributing features influencing the model's predictions.
- **Probability distribution** of Class 1 (smokers) shows consistent model prediction tendencies.

CONCLUSION and FUTURE SCOPE

Summarize Key Findings

- The Proposed Blending Model (**PBM**) combining **ESN + GoogleNet + AlexNet** achieved **~92% accuracy** in predicting smoking status.
- **Random Forest** feature selection and **Borderline-SMOTE** significantly improved model performance and stability.
- **SHAP explainability** provided clear insight into the most influential biosignal features affecting prediction.

Significance of Results

- Enables **early identification of hidden smokers**, improving cardiovascular disease (CVD) risk assessment.
- Supports **clinical decision-making** by providing **interpretable** model outputs.
- Demonstrates the effectiveness of integrating **temporal + spatial feature learning** for healthcare prediction tasks.

Future Scope

- Deploy PBM in **real-time wearable or hospital monitoring systems**.
- Expand dataset with **additional physiological signals** to enhance generalization.
- Perform **threshold tuning** to improve model confidence and reduce borderline prediction overlap.

Limitations

- Model performance may vary with **different population groups**
- Prediction probability distribution shows **tight clustering**, indicating need for further **decision boundary optimization.**

REFERENCES

- Ammar M. et al., "Hybrid deep learning model combining Echo State Networks and convolutional layers for detecting hidden smokers," *Proc. Intl. Conf. on Predictive Healthcare*, 2024.
- Odhiambo P., Singh R., and Lee T., "Gesture-based smoking prediction using smartwatch sensors and LSTM," *IEEE J. Biomedical Health Informatics*, 2022.
- Trigka K. and Dritsas S., "Improved SMOTE based on feature correlation for heart disease detection," *Computers in Biology and Medicine*, 2025.
- Hellen A. and Marvin D., "Interpretable smoking detection using VGG-16 and LRP," *Proc. IEEE ICCV*, 2024.
- Palanivel P. et al., "Smoking status prediction using biosignals with autoencoders and neural networks," *IEEE Access*, 2025.
- Pant A., Joshi M., and Kulkarni S., "Smoking prediction for COPD using residual neural networks," *IEEE Healthcom*, 2025.
- Yu L. et al., "Real-time smoking detection from ECG and GPS using BiLSTM networks," *IEEE Internet of Things Journal*, 2024.

QUESTIONS and ANSWERS

- Open the floor for questions from the audience

NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

ACKNOWLEGEMENTS

- We sincerely express our gratitude for the opportunity to present our project
- on smoking status prediction using the Proposed Blending Model (PBM).
- We thank the review panel and faculty members for dedicating their time to
- evaluate our work and for providing valuable suggestions that will help us
- improve this research further.

- For any queries or further discussion regarding this project, please contact:
- Email: duggisampath@gmail.com
- Phone: 8374204134